

Problem 1

Consider the sequence of functions

$$f_n(x) = n^2 x(1-x)^n, \quad x \in [0, 1].$$

(a) Pointwise convergence:

At the endpoints:

$$f_n(0) = n^2 \cdot 0 \cdot (1-0)^n = 0, \quad f_n(1) = n^2 \cdot 1 \cdot (1-1)^n = 0.$$

For $x \in (0, 1)$, let $r = 1 - x \in (0, 1)$. Then

$$f_n(x) = n^2 x r^n.$$

Consider

$$\ln f_n(x) = \ln(n^2 x r^n) = \ln x + 2 \ln n + n \ln r.$$

Since $0 < r < 1$, $\ln r < 0$, so

$$\lim_{n \rightarrow \infty} \ln f_n(x) = \lim_{n \rightarrow \infty} (\ln x + 2 \ln n + n \ln r) = -\infty.$$

Therefore,

$$\lim_{n \rightarrow \infty} f_n(x) = e^{-\infty} = 0.$$

Combining all cases, $f_n(x) \rightarrow 0$ for all $x \in [0, 1]$. Hence, the pointwise limit function is

$$f(x) = 0.$$

(b) Limit of the integrals:

$$\int_0^1 f_n(x) dx = \int_0^1 n^2 x(1-x)^n dx.$$

Use the substitution $u = 1 - x$, $du = -dx$, $x = 1 - u$:

$$\int_0^1 n^2 x(1-x)^n dx = \int_1^0 n^2(1-u)u^n(-du) = \int_0^1 n^2(1-u)u^n du.$$

Split the integral:

$$\int_0^1 n^2(1-u)u^n du = n^2 \int_0^1 u^n du - n^2 \int_0^1 u^{n+1} du = n^2 \left(\frac{1}{n+1} - \frac{1}{n+2} \right).$$

Simplify:

$$n^2 \left(\frac{1}{n+1} - \frac{1}{n+2} \right) = n^2 \cdot \frac{(n+2) - (n+1)}{(n+1)(n+2)} = \frac{n^2}{(n+1)(n+2)}.$$

Take the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \lim_{n \rightarrow \infty} \frac{n^2}{n^2 + 3n + 2} = 1.$$

(c) Uniform convergence:

The pointwise limit function is $f(x) = 0$, so

$$\int_0^1 f(x) dx = 0.$$

However,

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = 1 \neq 0.$$

If f_n converged uniformly to f on $[0, 1]$, we would have

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx = \int_0^1 f(x) dx = 0,$$

which is a contradiction. Therefore, f_n does **not** converge uniformly to f on $[0, 1]$.

Problem 2

Consider the sequences of functions

$$f_n(x) = \sqrt{x^2 + \frac{1}{n}}, \quad g_n(x) = \frac{\sin(nx)}{n^2}, \quad x \in [0, 1].$$

(a) Pointwise convergence:

For f_n :

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \sqrt{x^2 + \frac{1}{n}} = \sqrt{x^2} = |x| = x, \quad x \in [0, 1].$$

Thus, the limit function is

$$f(x) = x.$$

For g_n , note that

$$-\frac{1}{n^2} \leq \frac{\sin(nx)}{n^2} \leq \frac{1}{n^2}, \quad \forall x \in [0, 1].$$

By the Squeeze Theorem, since

$$\lim_{n \rightarrow \infty} -\frac{1}{n^2} = 0 \quad \text{and} \quad \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0,$$

it follows that

$$\lim_{n \rightarrow \infty} g_n(x) = 0, \quad x \in [0, 1].$$

Hence, the limit function is

$$g(x) = 0.$$

(b) Derivatives of f_n and convergence:

$$f'_n(x) = \frac{x}{\sqrt{x^2 + 1/n}}.$$

- For $x > 0$:

$$\lim_{n \rightarrow \infty} f'_n(x) = \frac{x}{\sqrt{x^2}} = 1 = f'(x).$$

- At $x = 0$:

$$f'_n(0) = 0, \quad f'(0) = 1.$$

Therefore,

$$\lim_{n \rightarrow \infty} f'_n(x) = f'(x) \quad \text{for all } x > 0,$$

but the convergence fails at $x = 0$.

(c) Derivatives of g_n and convergence:

$$g'_n(x) = \frac{n \cos(nx)}{n^2} = \frac{\cos(nx)}{n}.$$

Since

$$-\frac{1}{n} \leq \frac{\cos(nx)}{n} \leq \frac{1}{n}, \quad \forall x \in [0, 1],$$

by the Squeeze Theorem,

$$\lim_{n \rightarrow \infty} g'_n(x) = 0 = g'(x), \quad x \in [0, 1].$$

Thus, the derivatives of g_n converge to g' uniformly on $[0, 1]$.

Problem 3

Consider the series of functions

$$\sum_{n=1}^{\infty} f_n(x), \quad f_n(x) = \frac{x^2 e^{-nx}}{n^2}, \quad x \in [0, \infty).$$

(a) Find the maximum $M_n = \sup_{x \in [0, \infty)} |f_n(x)|$:

Compute the derivative:

$$f'_n(x) = \frac{1}{n^2} (2x e^{-nx} + x^2 (-n) e^{-nx}) = \frac{1}{n^2} x e^{-nx} (2 - nx).$$

Set $f'_n(x) = 0$ to find critical points:

$$x = 0 \quad \text{or} \quad x = \frac{2}{n}.$$

Evaluate f_n at these points:

$$f_n(0) = 0, \quad f_n\left(\frac{2}{n}\right) = \frac{(2/n)^2 e^{-2}}{n^2} = \frac{4e^{-2}}{n^4}.$$

Thus, the maximum value is

$$M_n = \frac{4e^{-2}}{n^4}.$$

(b) Uniform convergence via the Weierstrass M -test:

For all $x \in [0, \infty)$ and all $n \in \mathbb{N}$,

$$|f_n(x)| \leq M_n = \frac{4e^{-2}}{n^4}.$$

Since

$$\sum_{n=1}^{\infty} M_n = 4e^{-2} \sum_{n=1}^{\infty} \frac{1}{n^4}$$

is a convergent p -series with $p = 4 > 1$, the Weierstrass M -test implies that

$$\sum_{n=1}^{\infty} f_n(x)$$

converges uniformly on $[0, \infty)$.

Problem 4

Let $f : [0, 2] \rightarrow \mathbb{R}$ be continuous on $[0, 2]$ and differentiable on $(0, 2)$, with $f(0) = 0$, $f(2) = 6$, and $f'(x) \leq 3$ for all $x \in (0, 2)$.

By the Mean Value Theorem, there exists $c \in (0, 2)$ such that

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{6}{2} = 3.$$

Now, since $f'(x) \leq 3$ for all $x \in (0, 2)$, we integrate over $[0, 2]$:

$$f(2) - f(0) = \int_0^2 f'(x) dx \leq \int_0^2 3 dx = 6.$$

But $f(2) - f(0) = 6$, so equality holds:

$$\int_0^2 f'(x) dx = \int_0^2 3 dx.$$

Since $f'(x) \leq 3$ for all x and the integrals are equal, it follows that

$$f'(x) = 3 \quad \text{for all } x \in (0, 2).$$

Therefore, f is linear with slope 3, so

$$f(x) = 3x + C.$$

Using $f(0) = 0$, we get $C = 0$. Hence,

$$\boxed{f(x) = 3x.}$$

Problem 5

Let $f(x) = \log x$. Then f is continuous on $(0, \infty)$ and differentiable on $(0, \infty)$, with

$$f'(x) = \frac{1}{x}.$$

Fix arbitrary $2 \leq x < y < \infty$. Since f is continuous on the closed interval $[x, y]$ and differentiable on the open interval (x, y) , the Mean Value Theorem applies. Hence, there exists $c \in (x, y)$ such that

$$\frac{\log y - \log x}{y - x} = f'(c) = \frac{1}{c}.$$

Since $c \in (x, y)$ and $x \geq 2$, we have $c \geq 2$, so

$$\frac{1}{c} \leq \frac{1}{2}.$$

Therefore,

$$\frac{\log y - \log x}{y - x} \leq \frac{1}{2},$$

which implies

$$\log y - \log x \leq \frac{1}{2}(y - x),$$

for all $2 \leq x < y < \infty$.

Problem 6

Let $f(x) = \sqrt{x} = x^{1/2}$.

(a) Compute derivatives:

$$f'(x) = \frac{1}{2}x^{-1/2}, \quad f''(x) = -\frac{1}{4}x^{-3/2}.$$

Evaluate at $a = 1$:

$$f(1) = 1, \quad f'(1) = \frac{1}{2}, \quad f''(1) = -\frac{1}{4}.$$

The second-order Taylor polynomial centered at $a = 1$ is

$$P_2(x) = f(1) + f'(1)(x - 1) + \frac{f''(1)}{2}(x - 1)^2.$$

Thus,

$$P_2(x) = 1 + \frac{1}{2}(x - 1) - \frac{1}{8}(x - 1)^2.$$

(b) Approximate $\sqrt{1.2}$ using P_2 :

$$P_2(1.2) = 1 + \frac{1}{2}(0.2) - \frac{1}{8}(0.2)^2 = 1 + 0.1 - 0.005 = 1.095.$$

(c) By the Taylor Remainder Theorem, there exists $\xi \in (1, 1.2)$ such that

$$R_2(1.2) = \frac{f^{(3)}(\xi)}{3!}(1.2 - 1)^3.$$

Compute the third derivative:

$$f^{(3)}(x) = \frac{3}{8}x^{-5/2}.$$

Since $x^{-5/2}$ is decreasing on $[1, 1.2]$, we have

$$|f^{(3)}(\xi)| \leq \frac{3}{8}.$$

Hence,

$$|R_2(1.2)| \leq \frac{1}{3!} \cdot \frac{3}{8} \cdot (0.2)^3 = \frac{3}{48} \cdot 0.008 = 0.0005.$$

Therefore,

$$|f(1.2) - P_2(1.2)| \leq 0.0005.$$